Experiment 9

PMOS Buck (Step-Down) Switching Regulator: 1.2 V Output from a 3 to 5 V Supply with Ripple of 0.1% or Less

Important: Every component value and every number in Section 8 (inductor and capacitor ESR and DCR, MOSFET R_DS(on), diode drop, gate charge, supply currents of the control circuit) is illustrative, chosen to show the method and the expected shape of the waveforms. Your inductors, capacitors, MOSFET and diode will differ, and so will your results. Redo the design calculations of Section 4 with the datasheet or measured values of the parts you are actually given. The ripple limit of 1.2 mV peak-to-peak is very small: on a breadboard it is set mostly by capacitor ESR, wiring and the way the scope probe is connected, not by the ideal formulas. Do not copy numbers from this manual into your report.

1. Objective

To design, build and test a switching (buck) regulator with the same specification as the linear regulator of Experiment 8: 1.2 V output from a 3 V to 5 V input at up to 100 mA, with output ripple limited to 0.1% of the output voltage (1.2 mV peak-to-peak). To measure its efficiency, line and load regulation and output ripple, and to compare the results with the linear regulator.

2. Background

In a buck converter the P-MOSFET switch Q1 connects the inductor L1 alternately to V_IN (switch on, for a time D x T) and to ground through the Schottky diode D1 (switch off). The inductor and capacitor average the pulse train to a DC output. In continuous conduction, with V_F the diode drop:

$$V_OUT = D \times V_IN \quad \textit{(ideal)}, \qquad D = (V_OUT + V_F) / (V_IN + V_F) \quad \textit{(with the diode drop)}$$

A pulse-width modulator sets D. The error amplifier (a 741 connected as an integrator) compares V_OUT with V_REF and produces a control voltage V_C. A comparator compares V_C with a 100 kHz ramp of peak-to-peak amplitude V_ramp. The switch is on while V_C is above the ramp, so D = V_C / V_ramp. If V_OUT falls, V_C rises, D rises and V_OUT recovers. Because the integrator has no DC limit on its gain, V_OUT is held equal to V_REF.

Inductor ripple current. While the switch is off, the inductor current falls at a rate of (V_OUT + V_F) / L1, so the peak-to-peak ripple current is:

$$dI = (V_OUT + V_F)(1 - D) / (L1 \times f_s)$$

It is largest at the highest V_IN, where D is smallest. When the load current is smaller than dI / 2, the inductor current reaches zero during the off time (discontinuous conduction) and the relation between D and V_OUT changes. The loop still regulates, but the ripple and efficiency differ.

Output ripple. The ripple current dI flows into the output capacitor, and the ripple voltage has a capacitive part and an ESR part:

$$dV = dI \times (ESR + 1 / (8 f_s C))$$

Two things are therefore needed for a 1.2 mV limit: a small ripple current dI, which means a large product L1 x f_s, and a very low capacitor ESR. With dI of 16 mA, the ESR plus the capacitive term must be below about 75 mOhm. A single 100 uF electrolytic capacitor typically has an ESR of 0.1 Ohm to 0.5 Ohm, which is too high, but ESR divides by the number of capacitors in parallel. This design therefore uses one LC stage with the output capacitor built from four or more capacitors in parallel (400 uF, ESR of 25 mOhm or less). If your capacitors are worse than this, a second small LC stage (for example 10 uH and 100 uF) placed after the capacitor attenuates the ripple at f_s by about Z_C / Z_L, a factor of a few hundred, and can be added as an extension.

Why a slow loop. The L1 and C network resonates at about 305 Hz. An integrator-only loop is stable if its crossover frequency is kept far below this resonance, so the compensator is deliberately slow (crossover of 15 to 26 Hz). The price is a slow transient response, which is part of what you will observe.

Gate drive. Q1 is a P-channel high-side switch: its source is tied to V_IN and it conducts when the gate is pulled below V_IN by more than the threshold. The gate is therefore driven between V_IN (switch off, V_GS = 0) and 0 V (switch on, V_GS = -V_IN, that is -3 V to -5 V). No gate supply above V_IN and no bootstrap circuit is needed, and the gate never sees more than V_IN. The comparator output is low while the switch is on, so D is the fraction of the period for which the gate is low. A 100 k Ohm resistor from the gate to V_IN keeps Q1 off whenever the control circuit is not powered.

3. Specification

Parameter	Requirement
Input voltage, V_IN	3 V to 5 V (same as Experiment 8)
Output voltage, V_OUT	1.2 V
Load current	Up to 100 mA (1 k Ohm bleeder always connected)
Output ripple	0.1% of V_OUT = 1.2 mV peak-to-peak or less, at all V_IN and at 100 mA
Topology	Buck (step-down) with P-MOSFET high-side switch and Schottky freewheel diode
Switching frequency	100 kHz, set by the external ramp

4. Design Calculations

Do these calculations with the values of your own parts. The table gives the expressions and the results for the illustrative values: f_s = 100 kHz, L1 = 680 uH with DCR of 0.8 Ohm, C = 400 uF (four 100 uF capacitors in parallel) with a total ESR of 25 mOhm, V_F = 0.3 V, V_ramp = 2 V, R1 = 47 k Ohm and Ci = 330 nF.

Quantity	Expression	Result (illustrative)
Duty cycle D	D = (V_OUT + V_F) / (V_IN + V_F), V_F = 0.3 V	0.45 at 3 V, 0.28 at 5 V (0.48 and 0.30 with losses)
Inductor ripple current	dI = (V_OUT + V_F)(1 - D) / (L1 x f_s), L1 = 680 uH	12 mA at 3 V, 16 mA at 5 V
Continuous-conduction limit	I_LOAD(min) = dI / 2	About 8 mA at 5 V
Output ripple	dV = dI x (ESR + 1 / (8 f_s C)), C = 400 uF, ESR = 25 mOhm	0.34 mV at 3 V, 0.44 mV at 5 V (about 0.03%)
Largest ESR that still meets 1.2 mV	ESR(max) = 1.2 mV / dI - 1 / (8 f_s C)	About 70 mOhm (at 5 V)
Low-frequency resonance	f0 = 1 / (2 pi sqrt(L1 C))	About 305 Hz
Loop crossover	f_c = V_IN / (2 pi V_ramp R1 Ci), V_ramp = 2 V, R1 Ci = 15.5 ms	About 15 Hz at 3 V, 26 Hz at 5 V
Control voltage V_C	V_C = D x V_ramp	0.97 V at 3 V, 0.60 V at 5 V

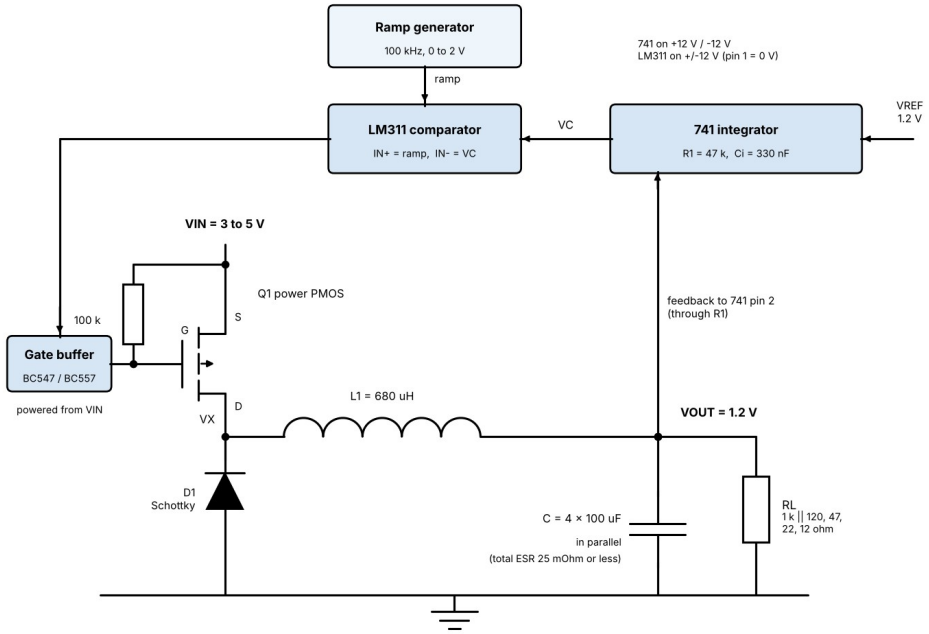
Choice of L1 and f_s. The ripple voltage is dI times the capacitor impedance at f_s, and dI is inversely proportional to L1 x f_s, so a larger L1 or a higher f_s directly reduces the ripple. 680 uH at 100 kHz gives a ripple current of only 12 mA to 16 mA, so the converter stays in continuous conduction down to about 8 mA. The cost of a larger inductor is its series resistance: a 680 uH, 300 mA part typically has a DCR of 0.5 Ohm to 0.8 Ohm, which drops about 80 mV and dissipates 8 mW at 100 mA. The loop corrects the drop automatically. A higher f_s would reduce the ripple instead, at the cost of higher switching and gate-drive loss.

Choice of the output capacitor. The capacitive part of the ripple, 1 / (8 f_s C), is only about 3 mOhm for 400 uF, so the ESR of the capacitor bank dominates. With 25 mOhm the calculated ripple is 0.34 mV to 0.44 mV, which is about three times below the 1.2 mV limit; the margin is needed because the real ripple on a breadboard also includes high-frequency spikes at the switching edges (from capacitor and wiring inductance) that this calculation ignores. Measure the ESR of your capacitors if you can (an LCR meter at 100 kHz) and use the largest-ESR formula in the table to check that the bank is good enough.

Choice of the compensator. With the integrator, the loop gain is V_IN / (V_ramp x s x R1 x Ci) times the response of the LC network. Setting the crossover frequency about ten times below the 305 Hz resonance keeps the loop stable whatever the damping. R3 = R1 is added in series with the non-inverting input so the 741 input bias current produces the same drop at both inputs. The remaining error is the input offset voltage (typically 1 mV) and offset current times 47 k Ohm (about 1 mV), which you can see in the V_OUT accuracy.

Estimate of the losses. The conduction loss in Q1, D1 and the inductor series resistances, the switching loss (about 0.5 x V_IN x I x (t_r + t_f) x f_s) and the gate-drive power (Q_g x V_IN x f_s, about 6 to 10 mW for a 20 nC gate, drawn from V_IN) are calculated for your parts and compared with the measurement in Part E.

5. Circuit and Components



Item	Value	Remarks
Switch Q1	Power P-MOSFET (lab stock)	Source to V_IN, drain to V_X. Must be fully on at V_GS = -3 V (see below and Section 6)
Gate pull-up	100 k Ohm, gate to V_IN	Keeps Q1 off when the control circuit is not powered
Freewheel diode D1	1N5819 Schottky	1 A, 40 V. Cathode to V_X, anode to ground
Inductor L1	680 uH	I_sat of 300 mA or more, DCR of 0.8 Ohm or less (80 mV drop at 100 mA)
Output capacitor C	4 x 100 uF low-ESR in parallel	Electrolytic or polymer, 10 V or more. Total ESR of 25 mOhm or less (each capacitor 0.1 Ohm or less). Several ceramics are even better. Mount at the load terminals
Error amplifier	741, R1 = 47 k, Ci = 330 nF	Integrator. Ci is a film or ceramic (NP0/X7R) capacitor. Pin 7 = +12 V, pin 4 = -12 V
Bias balance	R3 = 47 k, 100 nF to ground	In series with the 741 non-inverting input (pin 3), balances the input bias current
Reference	TL431 and pot, as in Experiment 8	Set V_REF to 1.200 V before connecting it
Comparator	LM311	Pin 8 = +12 V, pin 4 = -12 V, pin 1 = 0 V. IN+ (pin 2) = ramp, IN- (pin 3) = V_C, so the output is low (Q1 on) while V_C is above the ramp. Swap the two inputs if D falls when V_C is raised

Output pull-up	2.2 k Ohm to V_IN	On the LM311 open-collector output (pin 7)
Gate buffer	BC547 (NPN) and BC557 (PNP)	Complementary emitter followers: bases to pin 7, collectors to V_IN and ground, emitters joined to the gate
Ramp source	Function generator	Ramp or triangle, 100 kHz, 2 V peak-to-peak, offset of +1 V (so the ramp runs 0 V to 2 V)
Loads	1 k Ohm bleeder plus 120, 47, 22, 12 Ohm	As in Experiment 8, 0.5 W or more
Supplies	V_IN 3 to 5 V; +12 V and -12 V	Common ground. V_IN current limit 200 mA. The gate drive is powered from V_IN

Choosing the P-MOSFET. The on-state gate drive is the main constraint. At V_IN = 3 V the gate is only 3 V below the source, and many standard power PMOS devices have a threshold of -2 V to -4 V and are not fully on at that drive. Use a logic-level P-channel device, or check that R_DS(on) is specified at V_GS = -2.5 V or -4.5 V on the datasheet, and note its R_DS(on), V_GS(th), gate charge Q_g and maximum drain current. As a test, at V_IN = 3 V and 100 mA measure the voltage across Q1 when it is on: it should be 0.1 V or less. If it is larger, that device is not suitable for a 3 V input.

6. Safety Limits and Precautions

- Check your P-MOSFET datasheet before building: V_GS(th), R_DS(on) at V_GS = -2.5 V or -4.5 V, and V_DS(max) (at least 20 V). The gate is only ever driven between 0 V and V_IN (5 V at most), so the gate rating is not a concern, but the device must be well enhanced at V_GS = -3 V (see Section 5). Keep the 100 k Ohm gate pull-up fitted, otherwise a floating gate can turn Q1 on when V_IN is applied.

- Keep the V_IN current limit at 200 mA and never exceed 5 V. The load current must not exceed 100 mA. Do not connect the 12 Ohm load while the loop is open (Part B uses 22 Ohm).

- Never open the feedback path while V_IN is applied. With the feedback open the integrator can drive D to 100% and V_OUT rises to V_IN, which is more than 1.2 V and may damage whatever is connected to the output.

- Power-up order: first the +12 V and -12 V rails, then confirm that the ramp is 0 V to 2 V at 100 kHz and that V_REF = 1.200 V, then V_IN. Switch off in the reverse order.

- L1 must not saturate: choose an inductor rated 300 mA or more. A saturated inductor shows a spike in the current and gets hot.

- Electrolytic and tantalum capacitors are polarised and must be connected with the correct polarity, with a voltage rating of at least 10 V.

- Measuring a ripple of about 1 mV: use a short ground spring on the probe (not the long ground lead), AC coupling, 20 MHz bandwidth limit and 1 or 2 mV/div, and measure at the capacitor terminals. Before each measurement, touch the

probe tip to the ground spring at the same place to see the noise floor of the set-up.

- If the switch node rings at the edges, a small gate resistor (about 10 Ohm) and a short loop between the diode, Q1 and the output capacitor reduce it. Add them only if you see the ringing.

- Q1 and D1 may be warm after a few minutes at full load. Do not touch them while measuring.

7. Procedure

Part A - Check the modulator (low V_IN, switch removed). Build the control circuit and the gate buffer. Disconnect the 741 output from the LM311 and apply V_C from a variable DC source or a pot. With V_IN off, set the ramp to 100 kHz, 0 V to 2 V. Because the gate buffer is powered from V_IN, apply a low V_IN of 3 V with Q1 and the inductor disconnected (or Q1 removed) for this check. Measure the gate waveform on the scope (it swings between 0 V and V_IN and is low while the switch is on) and the duty cycle, taken as the fraction of the period for which the gate is low, for V_C of 0.5 V to 1.0 V in 0.1 V steps, and fill in Table 1. Check that D increases with V_C; if not, swap the comparator inputs.

Part B - Power stage with fixed duty (open loop). Keep V_C fixed from the pot and connect R_L = 22 Ohm. Set V_IN to 3 V, then 4 V and 5 V, and for each choose V_C so that V_OUT is about 1.2 V. Record V_C, the measured D (from V_X) and V_OUT in Table 2, and calculate V_OUT = D (V_IN + V_F) - V_F for comparison. Sketch V_X and the inductor-related waveform and mark the on-time and the diode conduction time.

Part C - Closed loop: line and load regulation. Reconnect the 741 integrator to the comparator and connect the feedback from V_OUT to pin 2 through R1. With R_L = 22 Ohm, record V_OUT and V_C for V_IN from 3.0 V to 5.0 V in steps of 0.5 V (Table 3). Then at V_IN = 3 V and 5 V step the load through the bleeder only, 120, 47, 22 and 12 Ohm, record I_LOAD and V_OUT, and note which loads are in continuous or discontinuous conduction (Table 4). Calculate line regulation in mV/V and load regulation in mV/A.

Part D - Output ripple. At V_IN = 3 V, 4 V and 5 V with R_L = 12 Ohm (about 100 mA), and at V_IN = 5 V with R_L = 47 Ohm (about 25 mA), measure the ripple voltage at the output with the scope, together with the probe noise floor (Table 5). Calculate the ripple as a percentage of V_OUT and state whether it is 0.1% or less. If it is not, investigate in this order: ground-lead length (use the spring ground tip), location of the capacitors relative to the load, ESR of the capacitor bank (add more capacitors in parallel), then a larger L1. As a last resort, add the optional second LC stage.

Part E - Efficiency. With R_L = 12 Ohm, set V_IN to 3.0, 3.5, 4.0, 4.5 and 5.0 V. Record I_IN (from the V_IN supply) and I_LOAD, and calculate P_IN = V_IN x I_IN, P_OUT = V_OUT x I_LOAD and the power-stage efficiency P_OUT / P_IN. Then read the currents drawn from the +12 V and -12 V rails once (the gate drive and pull-up are powered from V_IN and are already included in I_IN) and calculate P_AUX = sum of rail voltage times rail current. Calculate the total efficiency P_OUT / (P_IN + P_AUX) at 3 V and 5 V and enter the results in Table 6. Compare with Table 4 of Experiment 8.

Part F - Transient behaviour (optional). At V_IN = 5 V, switch the load between the bleeder only (or 47 Ohm) and 12 Ohm and observe V_OUT on the scope in AC-coupled mode. Record the size of the excursion and the time to return to within 5 mV. Change V_IN suddenly by 1 V and repeat.

8. Reference Simulation (illustrative values only)

The figures below were obtained from a switching-level simulation of the circuit using the illustrative values of Section 4, with ESR and series resistance included (R_DS(on) of 0.5 Ohm for the P-MOSFET) and a closed-loop integrator. They are provided to show what to expect, not as predictions for your parts.

Figure 1 shows the inductor current, the output ripple against the +/-0.6 mV limit band, and the switch state for V_IN = 3 V and 5 V at about 100 mA. In the simulation D is 0.48 and 0.30, dI is 12 mA and 16 mA, and the output ripple is 0.30 mV and 0.43 mV peak-to-peak, which is about 0.03% of 1.2 V. The simulation uses R_DS(on) = 0.5 Ohm for Q1, a DCR of 0.8 Ohm and a capacitor ESR of 25 mOhm. It does not include capacitor or wiring inductance or the edge spikes they cause, so a real circuit will show more ripple than this, and the 1.2 mV limit has to be checked by measurement.

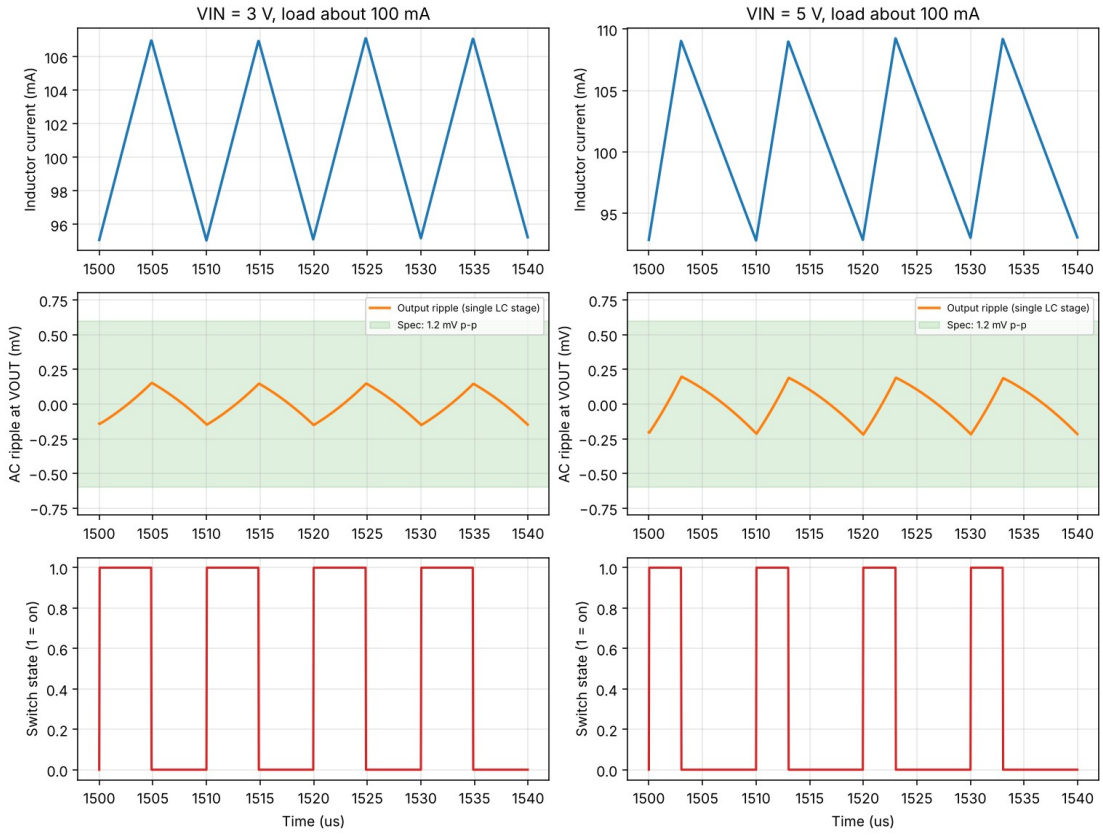


Figure 2 shows the closed-loop response. The loop settles without oscillation, but it is slow: a sudden 1 V step in V_IN (4 V to 5 V) causes an output excursion of about 0.36 V that recovers in about 35 ms, and a load change from 100 mA to 25 mA causes an excursion of about +116 mV that recovers in about 27 ms (about -108 mV and 18 ms for the opposite step). The excursions are large for such abrupt steps because of the low loop bandwidth chosen for stability; they are much smaller if V_IN or the load is changed slowly.

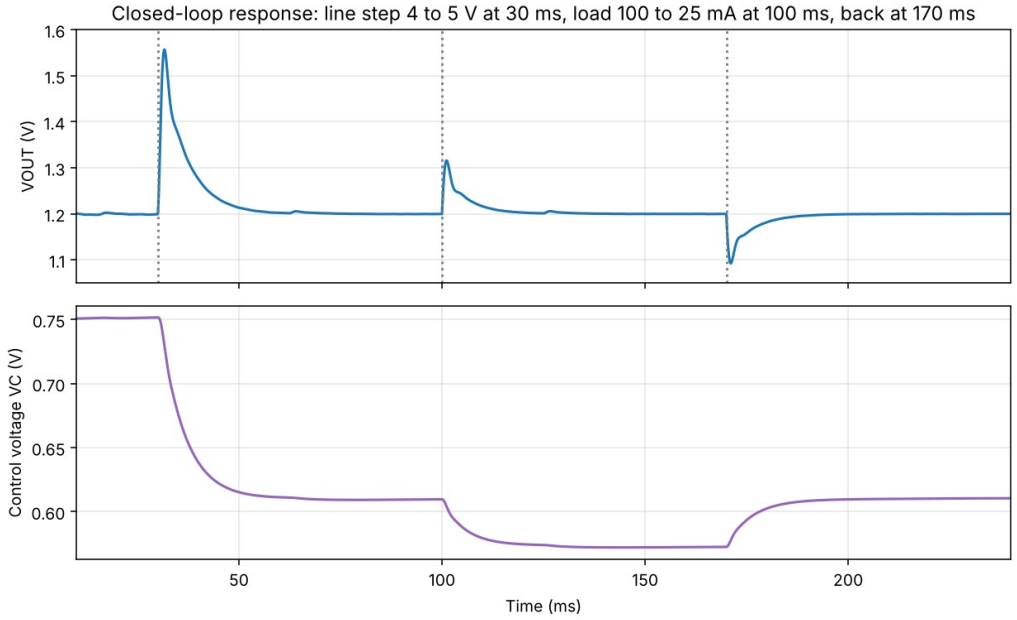
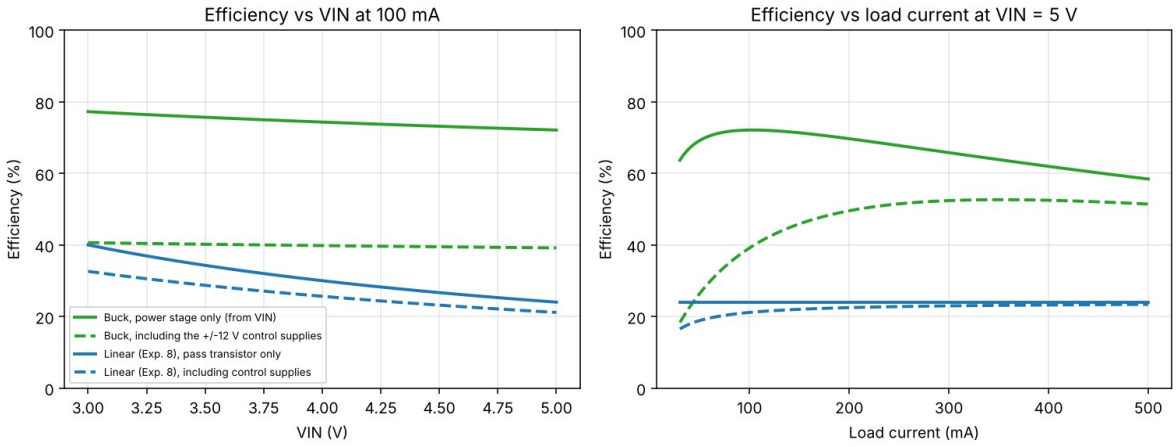


Figure 3 compares the efficiency of the buck regulator with that of the linear regulator of Experiment 8. The power-stage efficiency (counting only the power drawn from V_IN, including the gate drive) is about 77% at 3 V and 72% at 5 V at 100 mA, against 40% and 24% for the linear regulator, and the heat dissipated falls from 180 mW and 380 mW to about 35 mW and 46 mW. The dashed curves include the power drawn from the +12 V and -12 V rails (assumed 140 mW for the 741 and LM311 in the buck and 68 mW for the linear regulator). That overhead is larger than the 120 mW delivered at 100 mA, so with these discrete lab circuits the total efficiency is only about 41% and 39%, against 33% and 21% for the linear regulator. The overhead is fixed, so the advantage of the buck grows quickly with load current, as the right-hand plot shows.



9. Observation Tables (fill in using your own devices)

Table 1: Duty cycle of the modulator (V_IN off, ramp 0 V to 2 V, 100 kHz).

V_C (V)	D measured (%)	D predicted = V_C / 2 V (%)
0.50		25
0.60		30
0.70		35
0.80		40
0.90		45
1.00		50

Table 2: Open-loop power stage, R_L = 22 Ohm.

V_IN (V)	V_C set (V)	D measured (%)	V_OUT measured (V)	V_OUT predicted (V)
3.0				
4.0				
5.0				

Table 3: Closed-loop line regulation, R_L = 22 Ohm.

V_IN (V)	V_OUT (V)	V_C (V)
3.0		
3.5		
4.0		
4.5		
5.0		

Table 4: Closed-loop load regulation.

R_L (nominal)	I_LOAD (mA)	V_OUT (V) at V_IN = 3 V	V_OUT (V) at V_IN = 5 V	Mode (CCM or DCM)
Bleeder only				
120 Ohm				
47 Ohm				
22 Ohm				
12 Ohm				

Table 5: Output ripple.

V_IN (V)	I_LOAD (mA)	Ripple at V_OUT (mV p-p)	Probe noise floor (mV p-p)	Ripple (% of V_OUT)	Meets 0.1%?
3.0	100				
4.0	100				
5.0	100				
5.0	25				

Table 6: Efficiency at R_L = 12 Ohm (about 100 mA).

V_IN (V)	I_IN (mA)	P_IN (mW)	P_OUT (mW)	eta, power stage (%)	P_AUX (mW)	eta, total (%)
3.0						
3.5						
4.0						
4.5						
5.0						

Auxiliary-rail readings (record once): I(+12 V) = _____ mA, I(-12 V) = _____ mA, P_AUX = _____ mW. Total efficiency at V_IN = 3 V: _____ %, at V_IN = 5 V: _____ %.

Table 7: Comparison with the linear regulator at 100 mA.

At the same V_IN and 100 mA	Linear regulator (Experiment 8)	Buck regulator (this experiment)
Heat in the series device	(V_IN - V_OUT) x I	Conduction and switching loss only
Efficiency, V_IN only	V_OUT / V_IN	Table 6, power stage
Efficiency, all supplies	Experiment 8, Table 4	Table 6, total
Ripple at V_OUT	Noise only	Table 5
Line regulation	Experiment 8, Table 1	Table 3
Number of supplies needed	V_IN, +12 V, -12 V	V_IN, +12 V, -12 V, ramp source

10. Analysis and Discussion Questions

- Compare the measured D and V_OUT of Table 2 with the values predicted from D (V_IN + V_F) - V_F. What explains the difference?

- What line regulation (mV/V) and load regulation (mV/A) did you measure? Compare them with Experiment 8 and say what limits each.

- What was your measured output ripple? Compare it with the calculation of Section 4 and with the 1.2 mV limit, and explain what dominated any difference (ESR, probe, edge spikes).

- What is the largest total ESR your capacitor bank could have and still meet 0.1% ripple? What would you gain and lose by using a second LC stage instead of a larger L1?

- Which of your loads were in discontinuous conduction? How did the V_X waveform, the ripple and the efficiency change when the load fell below dI / 2?

- What power-stage efficiency and total efficiency did you measure at 3 V and 5 V? Compare with Table 4 of Experiment 8, and explain why the total efficiency is much lower than the power-stage efficiency.

- Using your measurements, make a loss budget at 5 V and 100 mA: conduction loss in Q1, loss in D1, inductor series resistance, switching loss and gate drive. Which is largest, and how would a synchronous rectifier (a second MOSFET in place of D1) change it?

- What happens to the ripple and to the efficiency if f_s is doubled to 200 kHz with the same L1 and C? Give the expected change in dI and in the loss terms.

- The loop recovers from a load step in about 20 ms. Why was a low crossover frequency chosen? Name one change to the compensator, or to the modulator (for example making the ramp amplitude proportional to V_IN), that would improve the line-step response.

- Why can the gate of a P-channel high-side switch be driven from V_IN itself, whereas an N-channel high-side switch needs a gate voltage above V_IN? What is the penalty of the P-channel device at V_IN = 3 V, and how did you check that your device is suitable?

- For which loads and input voltages would you choose the linear regulator of Experiment 8 over this buck regulator, and for which the buck regulator?